

WINDOW FUNCTIONS BASICS



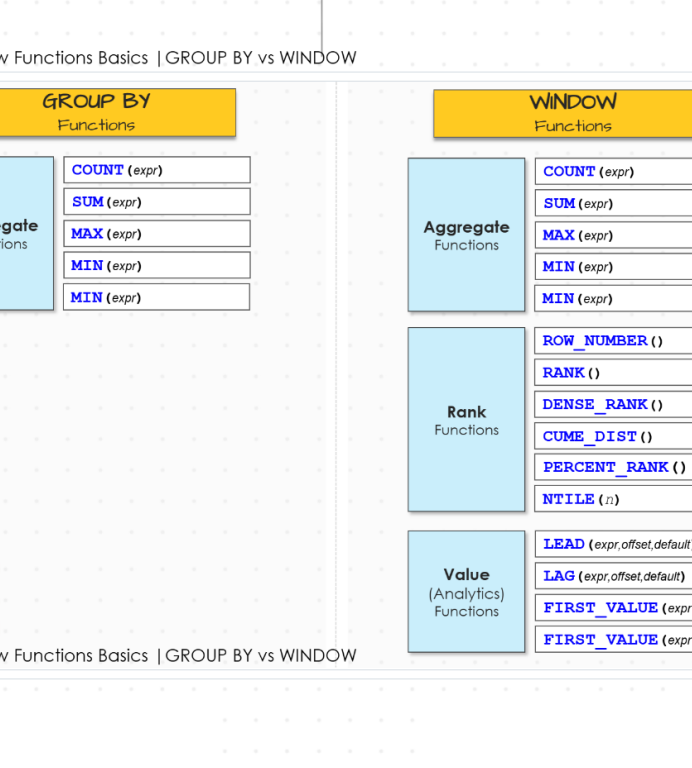
WINDOW FUNCTIONS

Perform calculations (e.g. aggregation) on a specific subset of data, without losing the level of details of rows.

SQL Course | Window Functions Basics

GROUP BY

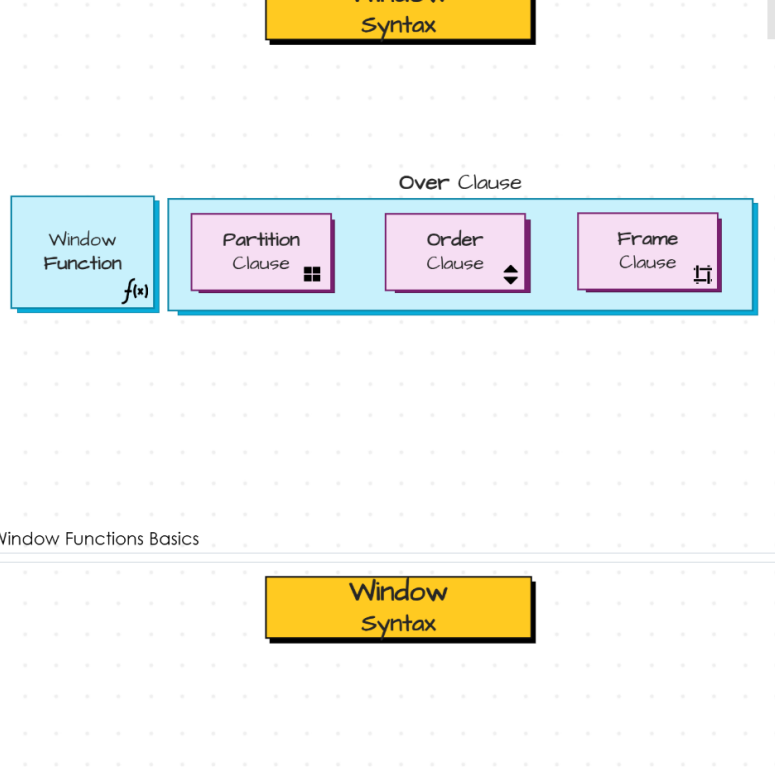
Aggregates and groups rows based on column/s into summary rows



SQL Course | Window Functions Basics | GROUP BY vs WINDOW

WINDOW Functions

Compute aggregates but keep details of individual rows at the same time



SQL Course | Window Functions Basics | GROUP BY vs WINDOW

GROUP BY Functions

Aggregate Functions	COUNT (expr)
	SUM (expr)
	MAX (expr)
	MIN (expr)
	MIN (expr)

Window Functions

Aggregate Functions	COUNT (expr)
	SUM (expr)
	MAX (expr)
	MIN (expr)
	MIN (expr)
Rank Functions	ROW_NUMBER ()
	RANK ()
	DENSE_RANK ()
	CUME_DIST ()
	PERCENT_RANK ()
Value (Analytic) Functions	NTILE (n)
	LEAD (expr,offset,default)
	LAG (expr,offset,default)
	FIRST_VALUE (expr)
	FIRST_VALUE (expr)

SQL Course | Window Functions Basics | GROUP BY vs WINDOW

GROUP BY

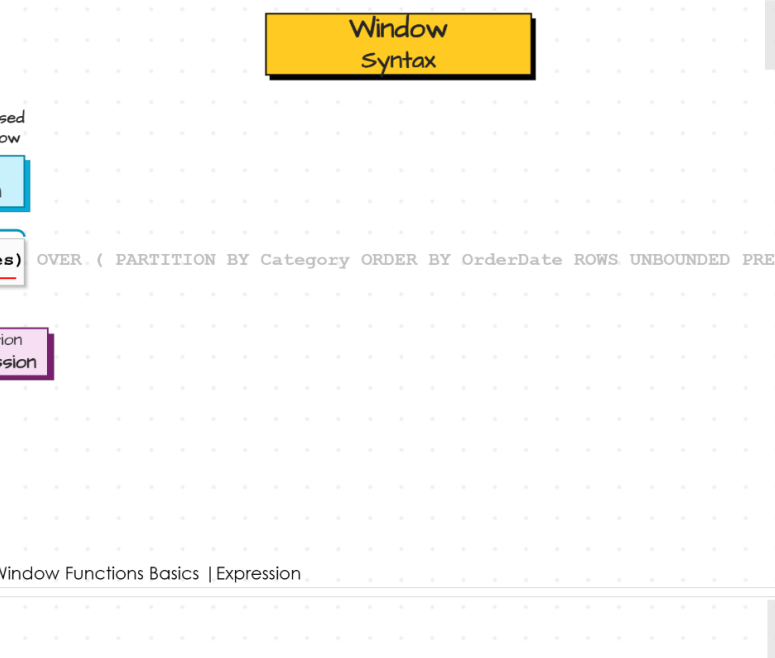
Simple Data Analysis (Aggregations)

WINDOW

Advanced Data Analysis (Aggregations + Details)

SQL Course | Window Functions Basics | GROUP BY vs WINDOW

Window Syntax



SQL Course | Window Functions Basics

Window Syntax

AVG(Sales) OVER (PARTITION BY Category ORDER BY OrderDate ROWS UNBOUNDED PRECEDING)

SQL Course | Window Functions Basics | Functions

Window Syntax

Calculation used on the Window f(x) Window Function AVG(Sales) OVER (PARTITION BY Category ORDER BY OrderDate ROWS UNBOUNDED PRECEDING)

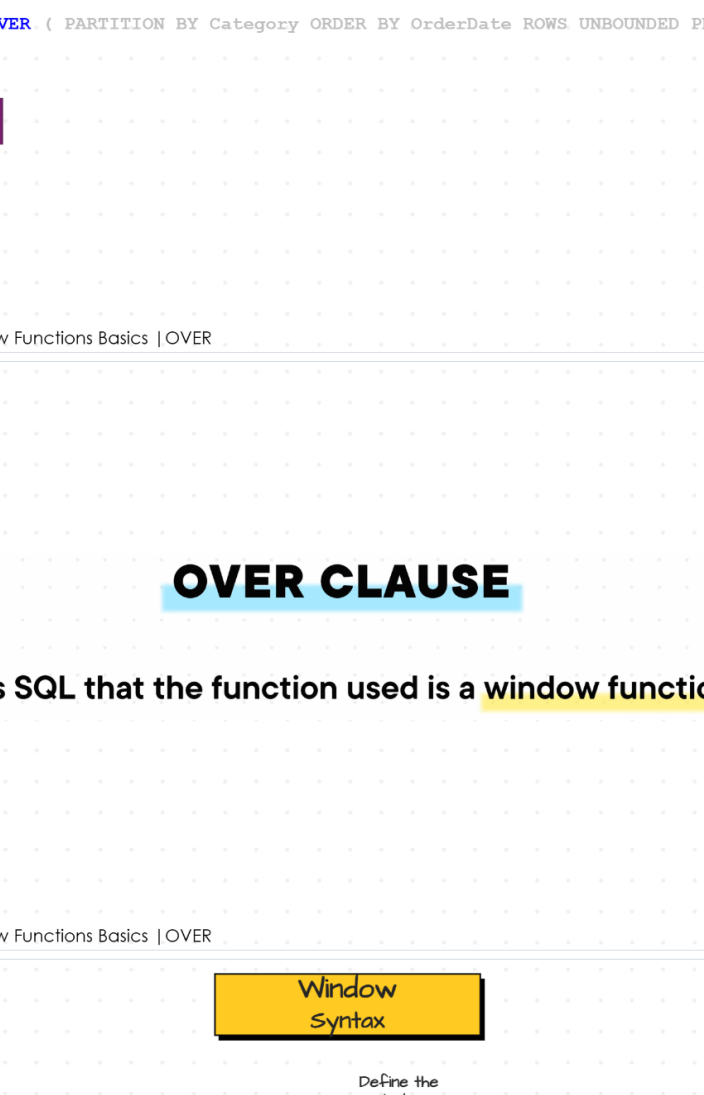
SQL Course | Window Functions Basics | Functions



WINDOW FUNCTIONS

Perform calculations within a window

SQL Course | Window Functions Basics | Functions



SQL Course | Window Functions Basics | Functions

Window Syntax

Calculation used on the Window f(x) Window Function AVG(Sales) OVER (PARTITION BY Category ORDER BY OrderDate ROWS UNBOUNDED PRECEDING)

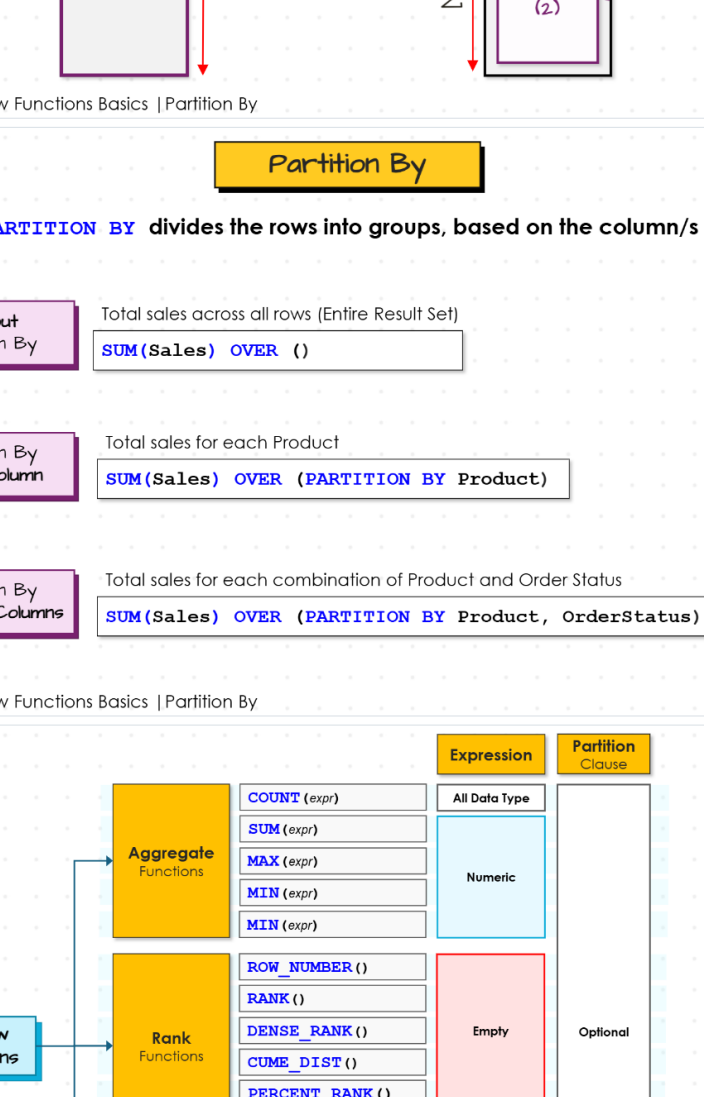
SQL Course | Window Functions Basics | Functions



WINDOW FUNCTIONS

Perform calculations within a window

SQL Course | Window Functions Basics | Functions



SQL Course | Window Functions Basics | Functions

Window Syntax

Calculation used on the Window f(x) Window Function AVG(Sales) OVER (PARTITION BY Category ORDER BY OrderDate ROWS UNBOUNDED PRECEDING)

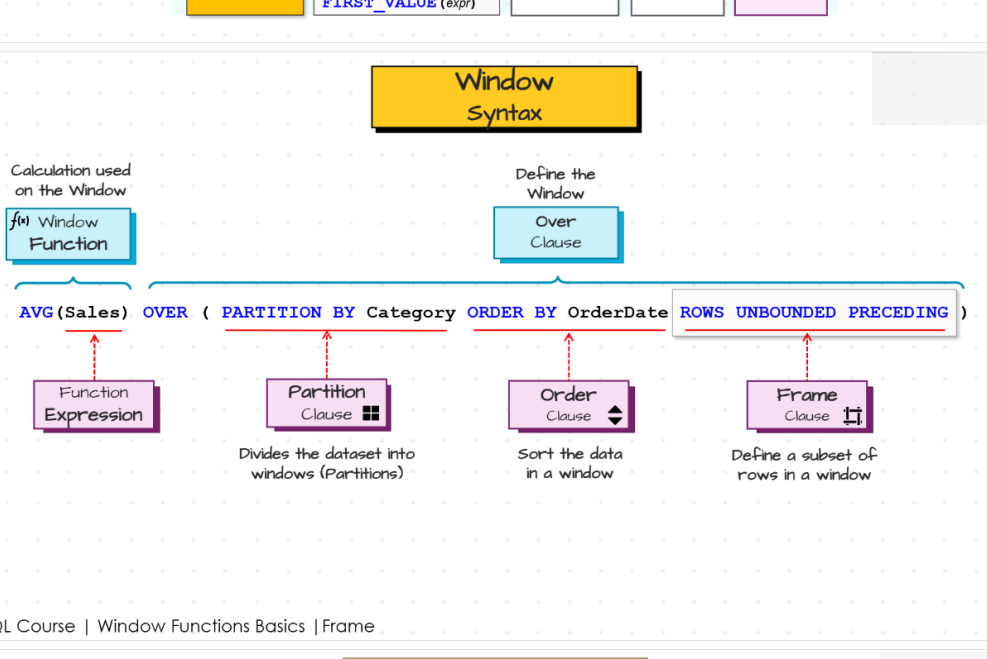
SQL Course | Window Functions Basics | Expression

FUNCTION EXPRESSION

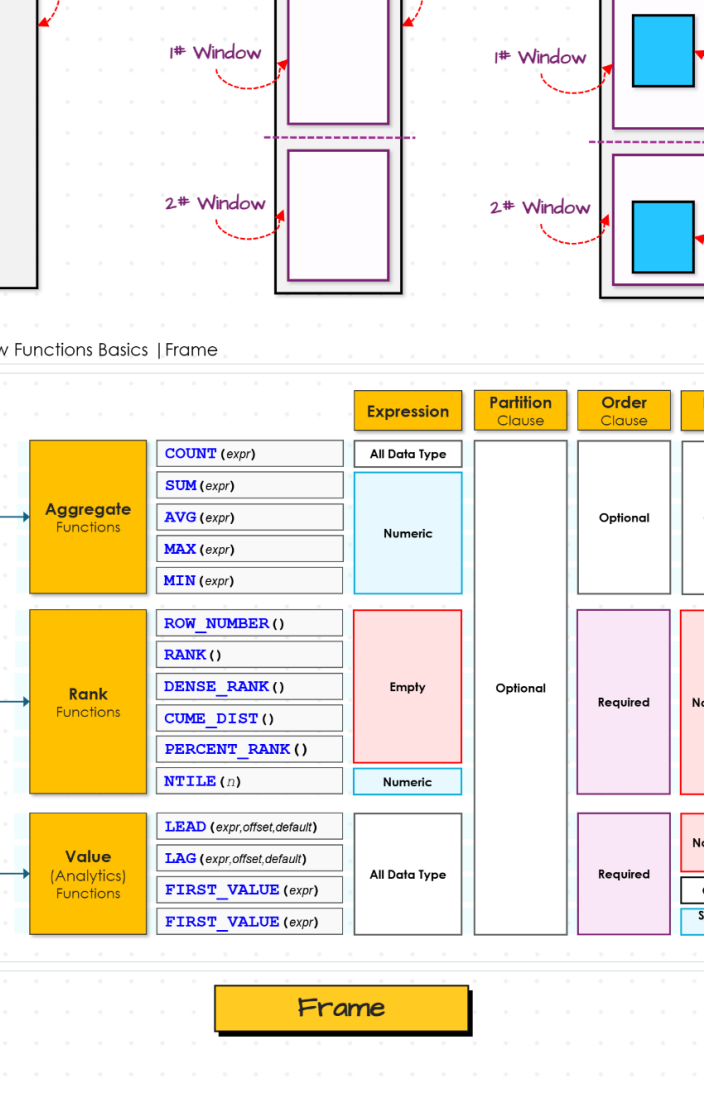
Arguments you pass to a function

SQL Course | Window Functions Basics | Expression

Window Expressions



SQL Course | Window Functions Basics | Expression



SQL Course | Window Functions Basics | Expression

Window Syntax

Calculation used on the Window f(x) Window Function AVG(Sales) OVER (PARTITION BY Category ORDER BY OrderDate ROWS UNBOUNDED PRECEDING)

SQL Course | Window Functions Basics | OVER

OVER CLAUSE

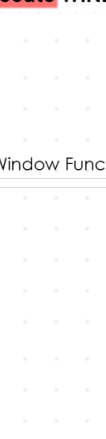
Tells SQL that the function used is a window function

SQL Course | Window Functions Basics | OVER

Window Syntax

Calculation used on the Window f(x) Window Function AVG(Sales) OVER (PARTITION BY Category ORDER BY OrderDate ROWS UNBOUNDED PRECEDING)

SQL Course | Window Functions Basics | Partition By



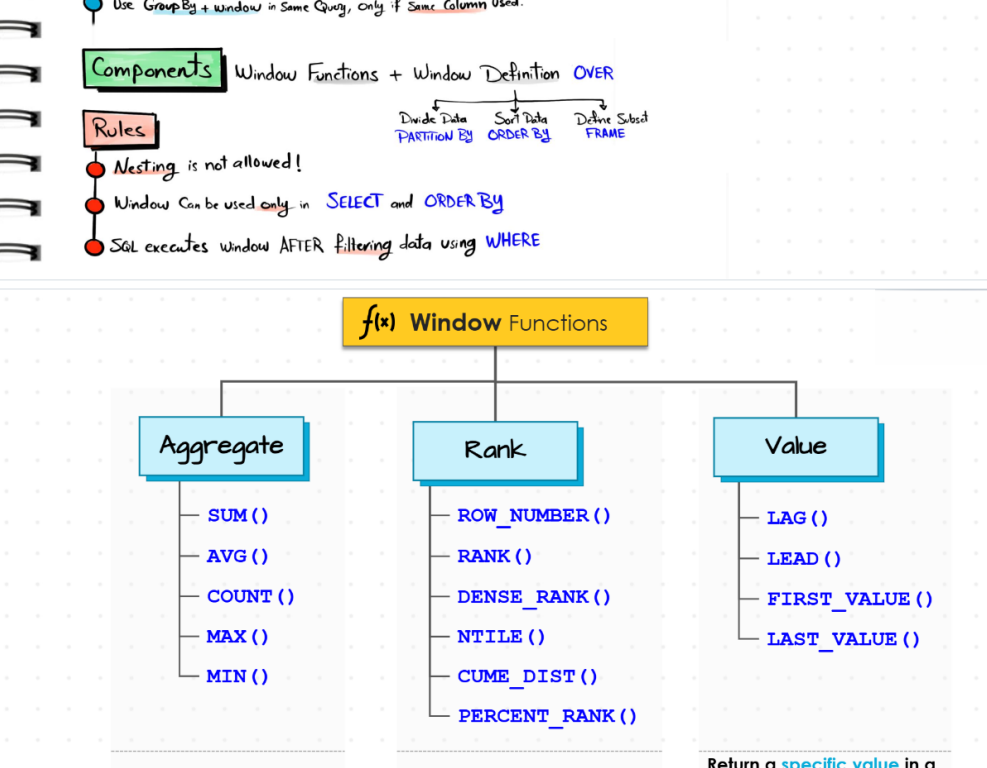
PARTITION BY

Divides the result set into partitions (Windows)

SQL Course | Window Functions Basics | Partition By

Partition By

PARTITION BY divides the rows into groups, based on the column/s



SQL Course | Window Functions Basics | Partition By

Partition By

PARTITION BY divides the rows into groups, based on the column/s

SQL Course | Window Functions Basics | Partition By

SQL Course | Window Functions Basics | Partition By

Window Syntax

Calculation used on the Window f(x) Window Function AVG(Sales) OVER (PARTITION BY Category ORDER BY OrderDate ROWS UNBOUNDED PRECEDING)

SQL Course | Window Functions Basics | Order By

Order By

Sort the data within a window (Ascending | Descending)

SQL Course | Window Functions Basics | Order By

SQL Course | Window Functions Basics | Order By

Window Syntax

Calculation used on the Window f(x) Window Function AVG(Sales) OVER (PARTITION BY Category ORDER BY OrderDate ROWS UNBOUNDED PRECEDING)

SQL Course | Window Functions Basics | Frame

Frame Clause

SQL Course | Window Functions Basics | Frame

SQL Course | Window Functions Basics | Frame

Frame

SQL Course | Window Functions Basics | Frame

Frame

SQL Course | Window Functions Basics | Frame

Frame

COMPACT FRAME

For only PRECEDING, the CURRENT ROW can be skipped

NORMAL FORM ROWS BETWEEN CURRENT ROW and 2 FOLLOWING

SHORT FORM ROWS 2 FOLLOWING

SQL Course | Window Functions Basics | Frame

Window Rules

#1 RULE

Window functions can be used ONLY in SELECT and ORDER BY Clauses

#2 RULE

Nesting Window Functions is not allowed!

#3 RULE

SQL execute WINDOW Functions after WHERE Clause

#4 RULE

Window Function can be used together with GROUP BY in the same query. ONLY if the same columns are used

SQL Course | Window Functions Basics | Rules

Window Rules

SQL Course | Window Functions Basics | Rules

